

Summary of Airbnb listings for Victoria (Vancouver, Canada)

Data Source: The data source of Airbnb listings for Victoria is an external source, collected from Airbnb website ([Get the Data | Inside Airbnb](#)) and not published by Airbnb itself. The website explicitly states that it neither endorses nor has any affiliation with Airbnb or its competitors. It emphasizes that the data is compiled solely for public analysis, discussion, and community engagement. In my view, this dataset can be considered reliable as it is sourced directly from Airbnb, and Inside Airbnb demonstrates a commendable level of transparency in presenting the data.

Data Collection: This dataset can be classified as usage data since it can be leveraged to optimize the pricing strategy for Airbnb listings. The data has been collected through web scraping, an automated process of extracting information from the Airbnb website. Given that the dataset contains listings before December 29, 2024, it is recent, resulting in very minimal time lag between data collection and its potential application. The dataset (*listings.csv*) contains information about Airbnb listings, including various details such as pricing, location, number of bedrooms, host information, and other relevant attributes.

A single record in this dataset represents one Airbnb listing available for rent. Each record contains information about the property, such as the listing's price, location, property type, room type, number of bedrooms, host information, availability, review scores, and other relevant details. This data helps analyze trends in pricing, demand, and availability across different neighborhoods in Victoria.

Data Limitations: The dataset is confined to information gathered on December 29, 2024, and does not account for any modifications to listings, prices, or reviews made after that date. Consequently, the data may not accurately reflect the current trends in the Victoria Airbnb market.

Why this Data set? This publicly available dataset is highly relevant to the country where I currently reside, offering valuable insights for optimizing Airbnb listing prices. It allows for an in-depth analysis of factors influencing the success of Airbnb listings. Additionally, the dataset presents an excellent opportunity to apply various machine learning algorithms, such as regression and classification, to extract meaningful patterns.

Data Profiling

1. Data Wrangling:

We have removed unnecessary columns from the data set.

```
[28]: # Removing columns that won't be useful for further analysis of the project
vic_listings = vic_listings.drop(columns = ['listing_url', 'picture_url', 'neighborhood_overview', 'last_scraped', 'source',
                                           'description', 'name', 'host_url', 'host_name', 'host_about', 'host_thumbnail_url',
                                           'host_location', 'host_picture_url', 'first_review', 'last_review',
                                           'host_neighbourhood', 'host_listings_count', 'host_total_listings_count',
                                           'host_verifications', 'host_has_profile_pic', 'minimum_minimum_nights',
                                           'maximum_minimum_nights', 'minimum_maximum_nights', 'maximum_maximum_nights',
                                           'minimum_nights_avg_ntm', 'maximum_nights_avg_ntm', 'license', 'calendar_updated',
                                           ])

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[41]: # Removing neighborhood column
vic_listings = vic_listings.drop(columns = ['neighbourhood', 'host_response_time', 'bathrooms_text'])
```

List of columns removed:

- **listing_url:** The URL link to the listing does not provide any analytical value for data analysis.
- **picture_url:** The picture URL of the listing is unnecessary as it does not contribute to measurable insights.
- **neighborhood_overview:** Contains unstructured text, making it difficult to analyze quantitatively.
- **last_scraped:** The date when the data was last updated is not useful for analysis purposes.
- **source:** The source of data (Airbnb) is constant and does not contribute any insights.
- **description:** The text-based listing description is not measurable for analysis purposes.
- **name:** The listing's name does not provide any meaningful data for analysis.
- **host_url:** The URL to the host's profile is irrelevant for data analysis.
- **host_name:** The host's name does not impact booking trends or pricing.
- **host_about:** Contains unstructured text and does not contribute to quantitative insights.
- **host_thumbnail_url:** The host's profile picture URL has no measurable impact on data analysis.

- **host_location:** The host's location may differ from the listing location, making it less useful.
- **host_picture_url:** Similar to the host_thumbnail_url, it does not provide analytical value.
- **first_review:** The date of the first review offers minimal insight into overall trends.
- **last_review:** The date of the last review is redundant since monthly review counts provide better insight.
- **host_neighbourhood:** The host's neighborhood is less relevant compared to the listing's neighborhood.
- **host_listings_count:** This information is duplicated by the host_total_listings_count column.
- **host_total_listings_count:** Duplicates host_listings_count, making it redundant.
- **host_verifications:** The verification methods are text-based and do not offer any measurable insight.
- **host_has_profile_pic:** Whether a host has a profile picture or not has no impact on booking trends.
- **minimum_minimum_nights:** Redundant as the minimum_nights column already captures the minimum stay requirement.
- **maximum_minimum_nights:** Adds no additional value since minimum_nights covers the same information.
- **minimum_maximum_nights:** Redundant because maximum_nights already captures maximum stay information.
- **maximum_maximum_nights:** Same as minimum_maximum_nights, making it unnecessary.
- **minimum_nights_avg_ntm:** Provides average minimum nights but does not contribute to core analysis insights.
- **maximum_nights_avg_ntm:** Similar to minimum_nights_avg_ntm, offering no additional insights.
- **license:** Licensing information is irrelevant for demand, pricing, or availability analysis.

- **calendar_updated:** The date of the calendar update does not contribute to measurable insights.
- **neighbourhood:** Replaced by neighbourhood_cleansed, which is more structured and 41.25% of data was missing (column was removed later during the analysis of the missing data).
- **host_response_time:** Already captured in the host_response_rate column, making it redundant (column was removed later during the analysis of the missing data).
- **bathrooms_text:** Provides text-based descriptions of bathrooms, already covered by the bathrooms column (column was removed later during the analysis of the missing data).

2. Missing Data

```
[38]: # Checking for missing values
vic_listings.isnull().sum()

[38]: id                                0
scrape_id                             0
host_id                               0
host_since                             0
host_response_time                     844
host_response_rate                     844
host_acceptance_rate                   398
host_is_superhost                      111
host_identity_verified                  0
neighbourhood                         1630
neighbourhood_cleansed                 0
neighbourhood_group_cleansed           0
latitude                               0
longitude                              0
property_type                          0
room_type                              0
accommodates                           0
bathrooms                             651
bathrooms_text                         1
bedrooms                              88
beds                                   652
amenities                              0
price                                  650
minimum_nights                         0
maximum_nights                         0
has_availability                       52
availability_30                        0
availability_60                        0
availability_90                        0
availability_365                       0
calendar_last_scraped                  0
number_of_reviews                      0
number_of_reviews_ltm                   0
number_of_reviews_l30d                  0
review_scores_rating                    477
review_scores_accuracy                  477
review_scores_cleanliness               477
review_scores_checkin                   477
review_scores_communication             477
review_scores_location                  477
review_scores_value                     477
instant_bookable                       0
listings_count                          0
listings_count_home                    0
listings_count_p_rooms                  0
listings_count_s_rooms                  0
reviews_per_month                       477
dtype: int64
```

The neighbourhood column has 41.25% missing data, so it will be dropped as the majority of its values are missing, making it less useful for analysis. The price column has 16.45%

missing data, which can be imputed using the mean based on relevant features. However, since the objective is to predict price using correlated features, removing rows with missing price values is a better approach to avoid introducing bias through imputation. It is important to note that this decision may lead to bias in the dataset, potentially impacting the accuracy of the results. Columns with less than 5% missing data will be retained since the amount of missing data is insignificant. For numerical columns with more than 5% missing data, imputation will be considered if it is necessary for the analysis. The columns `bathrooms_text` and `bathrooms` mean the same thing, so we will remove `bathrooms_text`. The `host_response_time` is not required for further analysis.

List of changes made:

- **Removing rows:** The price column has 16.45% missing data, which could be imputed using the mean value. However, since the primary objective is to predict price using correlated features, retaining imputed values may introduce bias and skew the results. Therefore, it was more appropriate to remove rows with missing price values to maintain data accuracy. It is important to acknowledge that this decision might still introduce bias by reducing the sample size, potentially impacting the overall accuracy of the analysis.
- **Removing columns:** The `neighbourhood` column was replaced by `neighbourhood_cleansed` due to the latter being more structured, and because 41.25% of the data in the original column was missing. The `host_response_time` column was removed as its information is already captured in the `host_response_rate` column, making it redundant. Additionally, the `bathrooms_text` column, which provided text-based descriptions of bathrooms, was eliminated as the `bathrooms` column already contained the necessary numerical data for analysis.
- **Formatting and Data Consistency Adjustments:** In the price column, currency symbols (\$) and commas (,) were removed to enable numeric analysis, and the values were then converted from string format to float for further aggregation. For the `host_acceptance_rate` and `host_response_rate` columns, the percentage symbol (%) was removed, and the values were converted to float for consistency.
- **Imputation of Missing Values:** Missing values in numerical columns such as `host_acceptance_rate`, `host_response_rate`, `review_scores`, `reviews_per_month`, and `price` were filled using the mean value of each respective column. This approach prevents data loss while ensuring the dataset remains consistent for analysis.
- **Mode Replacement:** For categorical columns like `bathrooms`, `bedrooms`, and `beds`, missing values were replaced with the most common value (mode), which was 1. This method ensures that the missing data does not skew the analysis or create

inconsistencies. Additionally, an outlier of 14 bedrooms was replaced with 1 bedroom to prevent distortion of the data.

- **Boolean Conversion:** Columns like `host_is_superhost`, `has_availability`, `host_identity_verified`, and `instant_bookable` had text values ('t' for true, 'f' for false). These were converted to binary numeric values (1 for true, 0 for false) to facilitate easier aggregation, statistical analysis, and visualizations. This conversion also standardized the dataset for advanced analytics.

3. Duplicate data:

```
[111]: vic_listings.duplicated().sum()  
[111]: 0
```

There are no duplicate entries in the dataset.

4. Checking for mixed data type

```
[154]: for col in vic_listings.columns.tolist():  
        weird = (vic_listings[col].applymap(type) != vic_listings[[col]].iloc[0].apply(type)).any(axis = 1)  
        if len (vic_listings[weird]) > 0:  
            print (col)
```

There are no mixed datatypes in the dataset.

Understanding the Data

```
[115]: # Descriptive statistics for data
vic_listings.describe()
```

	id	scrape_id	host_id	host_response_rate	host_acceptance_rate	latitude	longitude	accommodates	bathrooms	bedrooms	
count	3.301000e+03	3.301000e+03	3.301000e+03	3301.000000	3301.000000	3301.000000	3301.000000	3301.000000	3301.000000	3301.000000	3301.000000
mean	5.164819e+17	2.024123e+13	1.751980e+08	96.880282	87.101145	48.515786	-123.494869	3.812784	1.313996	1.567404	2.100000
std	5.071577e+17	1.844029e+00	1.838539e+08	13.685308	23.738052	0.151296	0.237684	2.261154	0.675052	1.023105	1.400000
min	4.116800e+04	2.024123e+13	1.329300e+04	0.000000	0.000000	48.328820	-124.424091	1.000000	0.000000	0.000000	0.000000
25%	3.385434e+07	2.024123e+13	2.858359e+07	100.000000	87.101145	48.423330	-123.521264	2.000000	1.000000	1.000000	1.000000
50%	6.170352e+17	2.024123e+13	9.488487e+07	100.000000	98.000000	48.451015	-123.412280	4.000000	1.000000	1.000000	2.000000
75%	9.900101e+17	2.024123e+13	2.972652e+08	100.000000	100.000000	48.554430	-123.364170	4.000000	1.500000	2.000000	3.000000
max	1.318260e+18	2.024123e+13	6.680625e+08	100.000000	100.000000	48.996710	-123.066340	16.000000	7.000000	7.000000	12.000000

Observation of the statistics for the dataset:

- The maximum price of 9999 is very high. It might be a big property or error in the pricing. So, we removed the rows where the price is 9956 and 9999 as they are clear outliers, and the hosts do not have any other listings to verify the pricing. Additionally, the prices appear excessively high compared to typical property sizes.
- The maximum value for minimum_nights and maximum_nights are 365 and 1125, which seems pretty high. We retained the row with minimum_nights set to 365 since the maximum_nights booked for that listing ID is 700, which exceeds the minimum nights requirement. Additionally, the maximum_nights count indicates that several bookings extend beyond 365 nights, suggesting long-term stays. There are no outliers in the maximum_nights column, as evidenced by the fact that 760 bookings were made for up to 1125 nights, indicating a consistent trend for extended stays.

Data Types

Variables	Data Types			
	Time-variant / -invariant	Structured / Unstructured	Qualitative / Quantitative	Qualitative: Nominal/Ordinal Quantitative: Discrete/Continuous
id	Time-invariant	Structured	Qualitative	Nominal
scrape_id	Time-invariant	Structured	Qualitative	Nominal
host_id	Time-invariant	Structured	Qualitative	Nominal
host_since	Time-invariant	Structured	Qualitative	Ordinal
host_response_rate	Time-variant	Structured	Quantitative	Continuous
host_acceptance_rate	Time-variant	Structured	Quantitative	Continuous
host_is_superhost	Time-variant	Structured	Quantitative	Continuous
host_identity_verified	Time-invariant	Structured	Qualitative	Nominal
neighbourhood_cleansed	Time-invariant	Structured	Qualitative	Nominal
neighbourhood_group_cleansed	Time-invariant	Structured	Qualitative	Nominal
latitude	Time-invariant	Structured	Quantitative	Discrete
longitude	Time-invariant	Structured	Quantitative	Discrete
property_type	Time-invariant	Structured	Qualitative	Nominal
room_type	Time-invariant	Structured	Qualitative	Nominal
accommodates	Time-variant	Structured	Quantitative	Discrete
bathrooms	Time-variant	Structured	Quantitative	Discrete
bedrooms	Time-variant	Structured	Quantitative	Discrete
beds	Time-variant	Structured	Quantitative	Discrete

amenities	Time-invariant	Structured	Qualitative	Ordinal
price	Time-variant	Structured	Quantitative	Discrete
minimum_nights	Time-variant	Structured	Quantitative	Discrete
maximum_nights	Time-variant	Structured	Quantitative	Discrete
has_availability	Time-variant	Structured	Quantitative	Discrete
availability_30	Time-variant	Structured	Quantitative	Discrete
availability_60	Time-variant	Structured	Quantitative	Discrete
availability_90	Time-variant	Structured	Quantitative	Discrete
availability_365	Time-variant	Structured	Quantitative	Discrete
calendar_last_scraped	Time-variant	Structured	Quantitative	Discrete
number_of_reviews	Time-variant	Structured	Quantitative	Discrete
number_of_reviews_ltm	Time-variant	Structured	Quantitative	Discrete
number_of_reviews_l30d	Time-variant	Structured	Quantitative	Discrete
review_scores_rating	Time-variant	Structured	Quantitative	Discrete
review_scores_accuracy	Time-variant	Structured	Quantitative	Discrete
review_scores_cleanliness	Time-variant	Structured	Quantitative	Discrete
review_scores_checkin	Time-variant	Structured	Quantitative	Discrete
review_scores_communication	Time-variant	Structured	Quantitative	Discrete
review_scores_value	Time-variant	Structured	Quantitative	Discrete
instant_bookable	Time-invariant	Structured	Qualitative	Nominal
listings_count	Time-variant	Structured	Quantitative	Discrete

listings_count_home	Time-variant	Structured	Quantitative	Discrete
listings_count_rooms	Time-variant	Structured	Quantitative	Discrete
listings_count_rooms	Time-variant	Structured	Quantitative	Discrete
reviews_per_month	Time-variant	Structured	Quantitative	Discrete

Continuous Variables to be used for further analysis:

- **Price:** It is crucial for analyzing price trends across different neighborhoods, property types, and amenities.
- **review_scores_rating:** This variable helps in understanding the overall guest satisfaction and its relationship with price or availability.
- **reviews_per_month:** Allows you to identify highly popular listings by seeing which ones receive more reviews monthly.

Categorical Variables likely to be used for further analysis:

- **neighbourhood_cleansed:** Helps in identifying which neighborhoods have the highest demand or best-rated listings.
- **property_type:** Allows comparison of demand, price, and reviews based on different property types (e.g., apartments, houses, etc.).
- **room_type:** Helps analyze the performance (such as price, reviews, availability) of different room types (such as Entire home, Private room, etc.).
- **host_is_superhost:** Useful for comparing booking rates and reviews between Superhosts and regular hosts.

Variables likely to be used for geographical information and analysis:

- **Latitude and Longitude:** To plot exact locations of Airbnb listings on a map.
- **Neighborhood and Neighborhood Group:** These columns allow to aggregate data based on regions such as most popular neighborhoods, highest-rated areas, etc.

Data Dictionary

Variable	Description	Variable Type
id	unique identifier for each listing	int64
scrape_id	unique identifier for each scraping session	int64
host_id	unique identifier for each host	int64
host_since	a member is a host from the year	object
host_response_rate	rate at which host response	float64
host_acceptance_rate	rate at which host accepts	float64
host_is_superhost	is a host good or not	object
host_identity_verified	is host identity verified or not	object
neighbourhood_cleansed	specific neighbourhood	object
neighbourhood_group_cleansed	specific neighbourhood	object
latitude	latitude of listing location	float64
longitude	longitude of listing location	float64
property_type	type of property (suite, townhouse, etc)	object
room_type	type of room (private room, entire home, etc.)	object
accommodates	number of people can stay	int64
bathrooms	no of bathrooms	float64
bedrooms	no of bedrooms	float64
beds	no of beds	float64
amenities	type of amenities with the property	object
price	price of booking	float64
minimum_nights	minimum nights booked	int64
maximum_nights	maximum nights booked	int64
has_availability	is there a availability or not?	object
availability_30	no of nights available in next 30 days	int64
availability_60	no of nights available in next 60 days	int64
availability_90	no of nights available in next 90 days	int64
availability_365	no of nights available in next 365 days	int64
calendar_last_scraped	most recent date and time when the availability calendar for a particular listing was scraped	object
number_of_reviews	no of reviews for listing	int64
number_of_reviews_ltm	no of reviews for listing in last 12 months	int64

number_of_reviews_l30d	no of reviews for listings in last 30 days	int64
review_scores_rating	overall rating given by guests	float64
review_scores_accuracy	reviews based on listing description	float64
review_scores_cleanliness	reviews based on property cleanliness	float64
review_scores_checkin	reviews based on checkin process	float64
review_scores_communication	reviews based on customer communication	float64
review_scores_value	reviews based on quality of stay	float64
instant_bookable	is the property instantly bookable or not?	object
listings_count	total no of listings	float64
listings_count_home	total no of entire home/apartment	int64
listings_count_private_rooms	total no of private rooms	int64
listings_count_shared_rooms	total no of shared rooms	int64
reviews_per_month	average no of reviews per month	float64

Questions to Explore

1. How are the listings geographically distributed in Victoria? Are there certain neighborhoods with a higher concentration of listings?
2. What is the distribution of property types and room types across Victoria's listings?
3. What are the most commonly offered amenities in the listings?
4. Is there a relationship between the number of bedrooms and the listing price?
5. Are certain neighborhoods associated with higher listing prices compared to others?
6. Does the type of property influence the listing price?
7. Is there any correlation between the number of accommodations offered and the listing price?
8. Which property types tend to have higher listing prices?
9. Is there a relationship between the number of reviews and the listing price?
10. Do customers tend to prefer listings with lower minimum night requirements or higher minimum night requirements?
11. Is there any correlation between the listing price and the average number of reviews per month?